

Submitted by

Adarsh Kumar Thakur (20BEC8007), Shubham Sharma (20BEC8004),
Abburi Hemant Naga Venkata Sai (20BEC8009), Sakshi Saini (20BEC8016),
Tenzin Sangay (19BEC1076)

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Dr. Manish Deshwal

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Abstract

Wheelchairs are vital mobility aids for individuals with lower body movement difficulties due to disease, injury, or congenital conditions. Technological advancements have introduced autonomous wheelchairs with adaptive features and IoT capabilities. However, most commercial electric models are expensive and unaffordable for many users.

This project introduces a cost-effective head-tilt controlled wheelchair suitable for paralyzed individuals or those with limited limb mobility. It uses an accelerometer sensor to detect head gestures for directional control. Arduino UNO boards serve as microcontrollers, with NRF24L01 modules enabling wireless communication. The system also incorporates provisions for health monitoring sensors (such as pulse sensors) and GSM/GPRS modules for real-time location and health data transmission to a server, accessible via a Google Assistant interface. This solution enhances user independence, safety, and remote monitoring capabilities.

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Chapter 1

Introduction

1.1 About the Project

The Internet of Things (IoT) connects physical devices embedded with electronics, software, sensors, and network connectivity, enabling them to collect and exchange data. This project develops an IoT-enabled wheelchair controlled by head movements. An accelerometer detects head tilts, Arduino microcontrollers process the signals, and NRF modules transmit commands wirelessly to control the wheelchair motors.

1.2 Motivation

With increasing cases of mobility disabilities, especially limb impairments, there is a strong need for intuitive and affordable assistive technologies. Traditional wheelchairs require manual effort, while high-end electric versions are costly. This head-tilt system provides easy control for users with paralysis. Integrated health sensors allow real-time monitoring, sending alerts in emergencies (e.g., falls, fainting) to family members via GSM, along with location tracking.

1.3 Objective

To design a cost-effective, IoT-based wheelchair operated via head gestures, integrated with health sensors and a Google Assistant bot for remote monitoring of health conditions and location.

1.4 Contribution

This project supports the “Make in India” initiative by utilizing locally available components. It empowers differently-abled individuals by providing an accessible, user-friendly mobility solution that does not require advanced technical skills. The system effectively merges modern electronics with practical engineering.

Chapter 2

Background Study

Wheelchairs assist people who have difficulty walking due to illness, injury, or disability. They can be manually or electrically propelled. As per the 2011 Census of India, a substantial percentage of the disabled population faces mobility issues.

Autonomous wheelchairs available in the market are often expensive and lack comprehensive emergency support features.

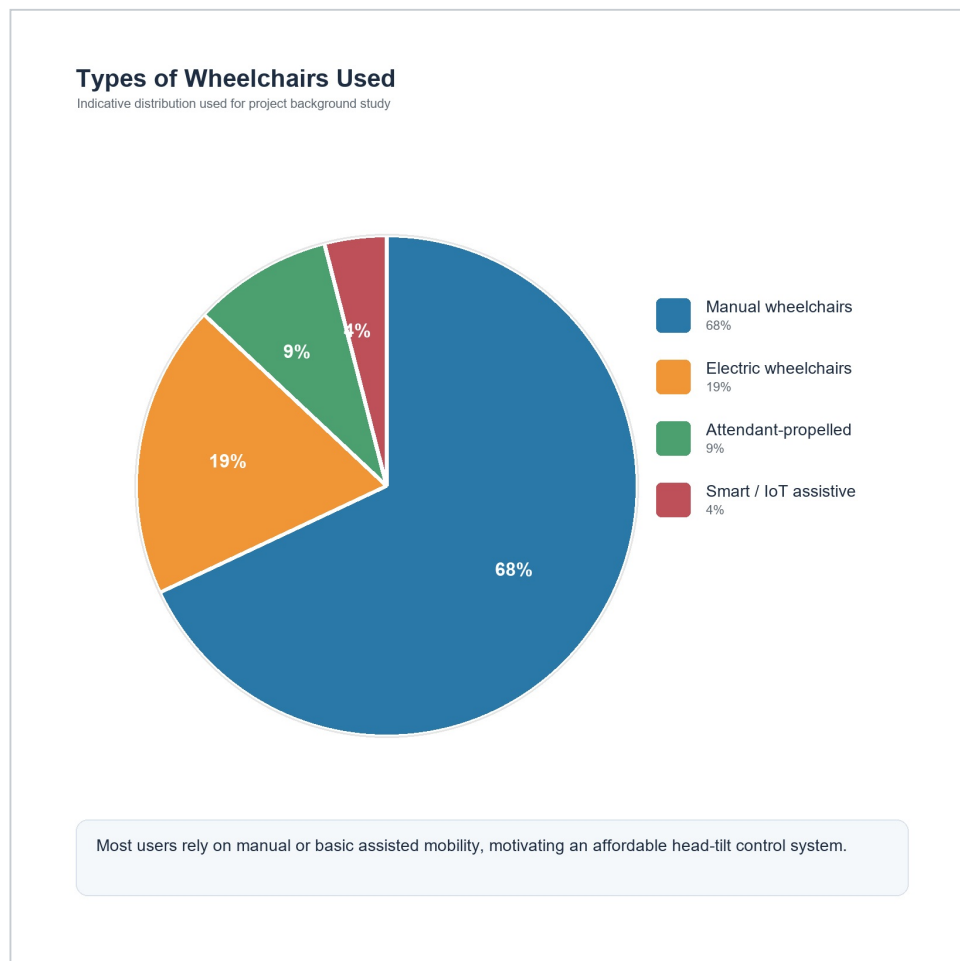


Figure 2.1: Pie chart indicating the type of wheelchairs used by people.

Voices | Comment

The true cost of being disabled goes far beyond just the physical

Disabled people are facing some of the toughest financial challenges

James Moore | Friday 25 July 2014



Article visual used for background study on mobility barriers and disability-related costs.

Figure 2.2: Article from web search about problems faced by handicapped people.

Arduino: Arduino is an open-source hardware and software platform used to create interactive projects. It converts code into physical actions. We utilize Arduino UNO boards for their ease of use.

Chapter 3

Requirement Analysis

3.1 Hardware Requirements

- **Arduino UNO (x2):** Open-source microcontroller board based on ATmega328P with 14 digital I/O pins and 6 analog inputs.
- **Accelerometer GY-61 (ADXL335):** Three-axis accelerometer for detecting head movements with low power consumption.
- **NRF24L01+ Modules:** Wireless communication modules operating at 2.4 GHz for data transmission between headband and wheelchair.
- **L293D Motor Driver IC:** Dual H-bridge driver for controlling DC motors.
- **Voltage Regulators (7805, 7809):** Provide stable 5V and 9V supplies.
- **Other Components:** Jumper wires, 12V rechargeable batteries, chassis, high-torque DC motors, headband.

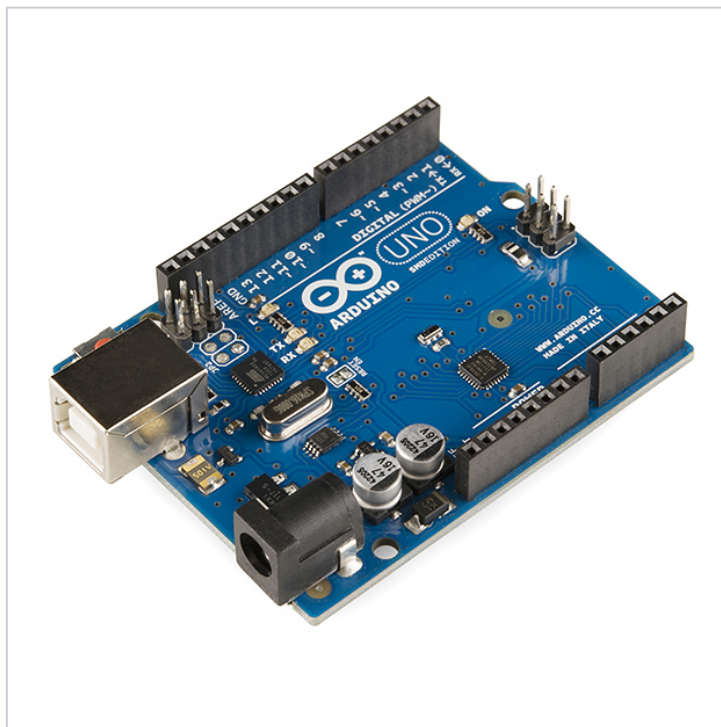


Figure 3.1: Arduino UNO Board

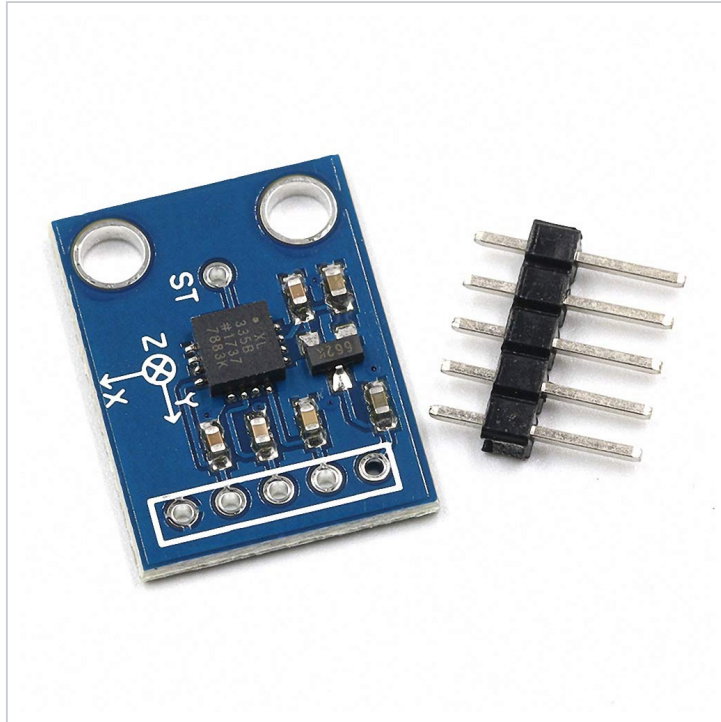


Figure 3.2: Accelerometer GY-61

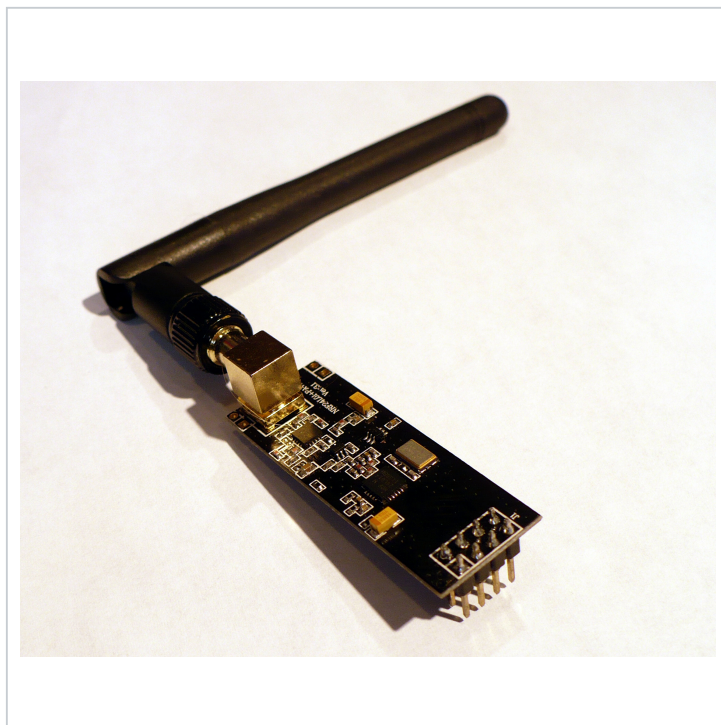


Figure 3.3: NRF24L01+ Module



Figure 3.4: L293D Motor Driver IC

3.2 Software Requirements

- Arduino IDE: For writing and uploading code to the microcontroller.
- Sublime Text: Advanced text editor for code development.

Chapter 4

Detailed Design

Detailed circuit diagrams were designed for both the headband and the wheelchair systems. The headband prototype was built using headphones, and the wheelchair base was implemented on a metal chassis.

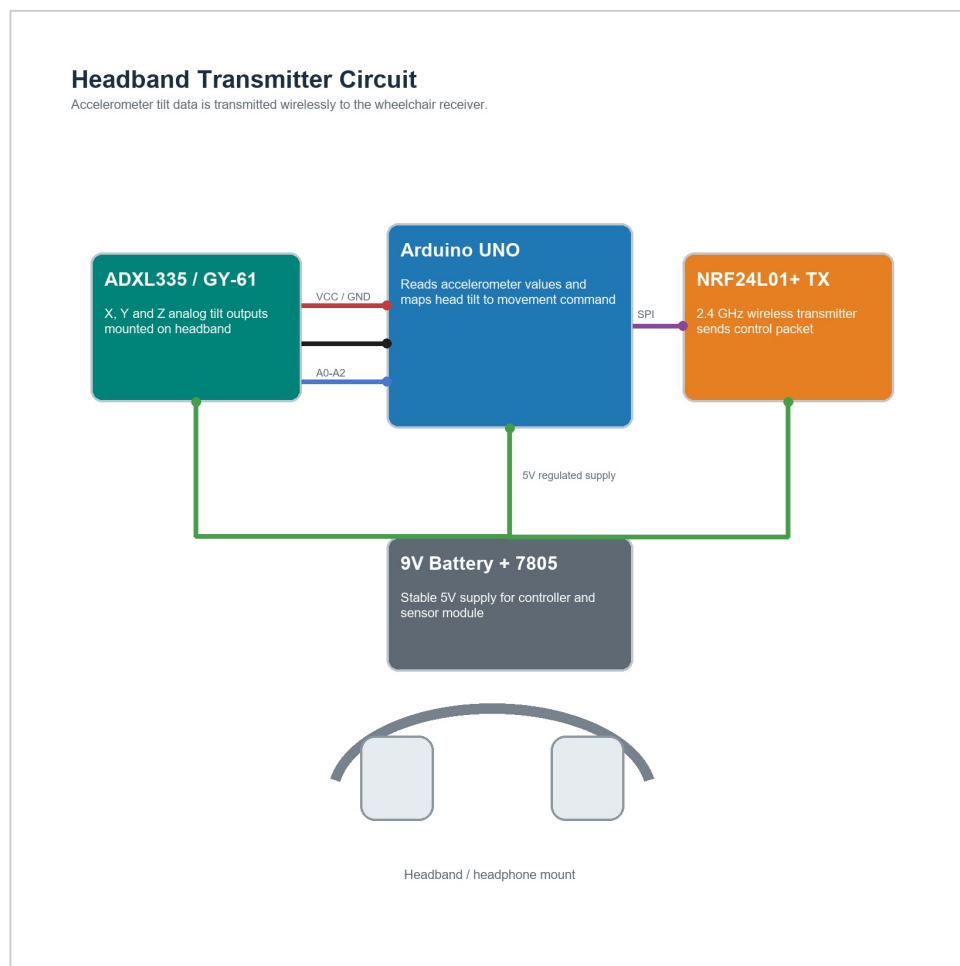


Figure 4.1: Circuit Diagram of Headband

Wheelchair Receiver Circuit

Receiver converts wireless commands into motor driver signals for left and right DC motors.

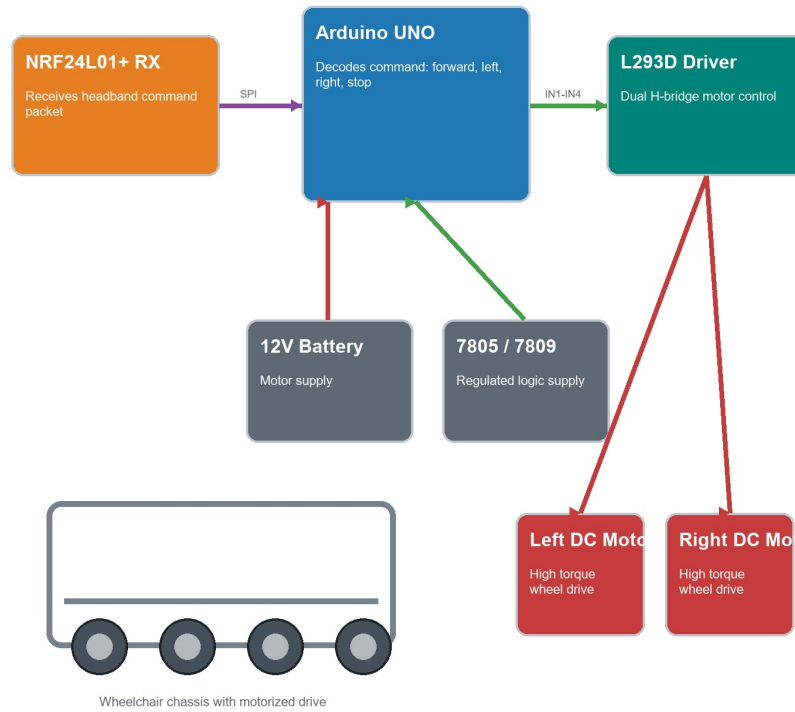


Figure 4.2: Wheelchair Circuit Diagram

Chapter 5

Implementation

The implementation is divided into core modules, with the head-movement control completed and health/IoT features planned.

Head Movement Control: The accelerometer on the headband maps tilt angles to directional commands (forward, left, right, stop). These commands are transmitted wirelessly via NRF modules to the wheelchair Arduino, which controls the motors through the L293D driver.

The system uses 12V batteries with voltage regulators for stable operation. The wheelchair is built on a chassis with two high-torque DC motors.

Chapter 6

Testing Reports

Testing was performed using the Arduino IDE Serial Monitor. Head tilts were verified to produce correct motor responses.

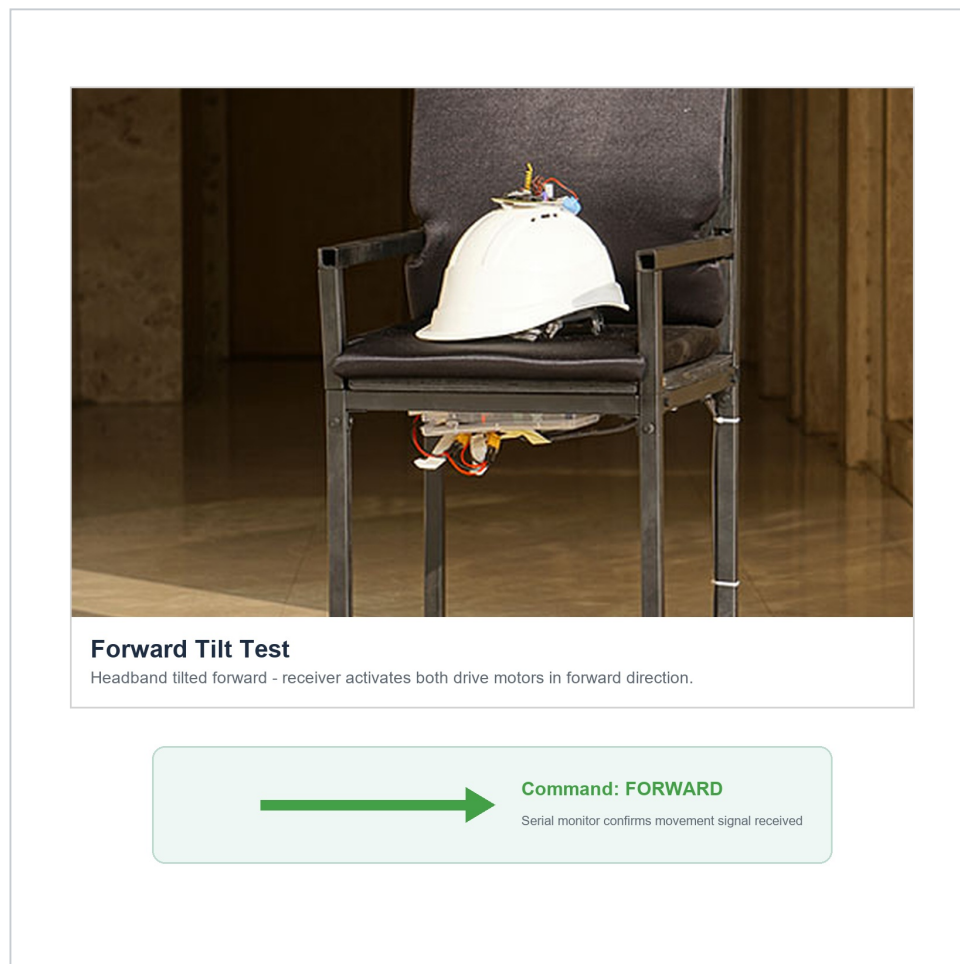
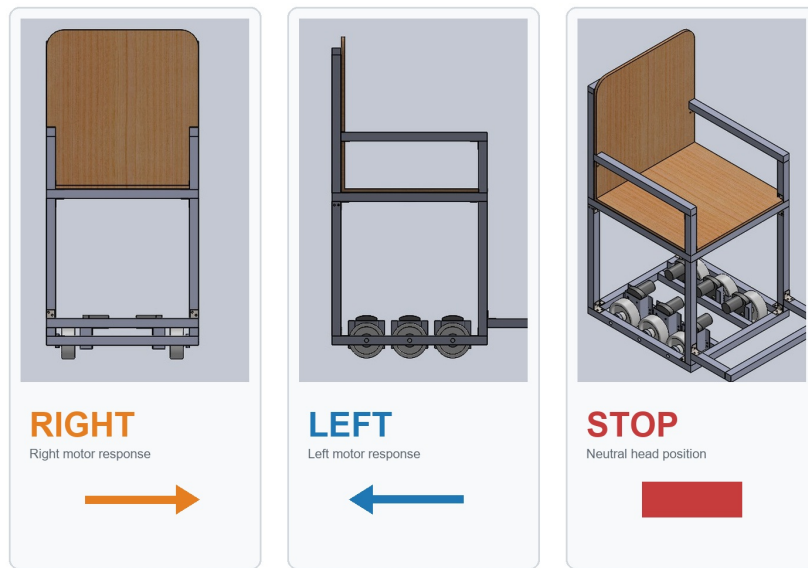


Figure 6.1: Head tilt Forward

Direction Testing: Right / Left / Stop

Wireless commands were checked against motor response for each head tilt condition.



Observed result:

Wheelchair responded to directional commands and stopped at neutral head position.

Figure 6.2: Head tilt Right / Left / Stop

Chapter 7

Conclusion

The Head Tilt Controlled Wheelchair provides an affordable and effective mobility solution for individuals with disabilities. Built using accessible components, it offers intuitive control and potential for health monitoring integration. Future enhancements include full GSM alerts, voice control, and machine learning features. This project significantly contributes to assistive technology and user independence.

Chapter 8

Proposed Future Work

Further development includes integrating health sensors, GSM/GPRS modules, Google Assistant integration, and a detailed Gantt chart for remaining tasks.

Chapter 9

References

1. James Moore, “The true cost of being disabled...”, The Independent.
2. Various research papers on smart wheelchairs and gesture control systems.
3. Component datasheets for Arduino, ADXL335, NRF24L01, etc.

Additional image sources

4. Arduino UNO photo: SparkFun Electronics via Wikimedia Commons, CC BY 2.0.
5. NRF24L01+ photo: Dmcole via Wikimedia Commons, CC BY-SA 3.0.
6. L293D photo: oomlout via Wikimedia Commons, CC BY-SA 2.0.
7. GY-61 ADXL335 module image: HiLetgo/Amazon image via Manuals+.
8. Head-tilt wheelchair prototype images: Nevon Projects.
9. Article visual: The Independent / Susannah Ireland. Pie chart and circuit diagrams created for this report.